

Research Sources

Annotated references supporting the Caffeine & Cardiovascular System guide.

SOURCE 01

Energy Drinks and Their Acute Effects on Heart Rhythm and Electrocardiographic Time Intervals in Healthy Children and Teenagers: A Randomized Trial

Authors: Guido Mandilaras, Pengzhu Li, Robert Dalla-Pozza, Nikolaus Alexander Haas, Felix Sebastian Oberhoffer

PMID: 35159306 · **PMCID:** PMC8834195 · **DOI:** 10.3390/cells11030498

Link: [View on PubMed →](#)

KEY FINDINGS

- Consumption of energy drinks led to a significant increase in supraventricular extrasystoles (SVES) — extra, abnormal heartbeats originating above the ventricles.
- The study observed a significant decrease in heart rate, which the authors attribute to a baroreflex response — the body's attempt to compensate for an acute spike in both systolic and diastolic blood pressure.
- There was a significant decrease in the QT–RR relationship (an electrocardiogram metric), indicating an autonomic nervous system response to the cardiovascular stress.

Conclusion: *Minors with underlying heart rhythm conditions could be at risk for malignant dysrhythmias after consuming these beverages.*

Relevance: *Provides strong evidence for both acute blood pressure elevation and arrhythmias. The observed heart rate decrease — caused by a sudden pressure spike — offers a nuanced talking point about how the cardiovascular system attempts to compensate for caffeine toxicity.*

SOURCE 02

Cardio- and Cerebrovascular Responses to the Energy Drink Red Bull in Young Adults: A Randomized Cross-Over Study

Authors: Erik K Grasser, Gayathri Yepuri, Abdul G Dulloo, Jean-Pierre Montani

PMID: 24474552 · **PMCID:** PMC4175045 · **DOI:** 10.1007/s00394-014-0661-8

Link: [View on PubMed →](#)

KEY FINDINGS

- Compared to water, Red Bull consumption significantly increased systolic and diastolic blood pressure, heart rate, and cardiac output.
- It increased the 'double product' (heart rate × systolic blood pressure), a direct measure of the workload and oxygen demand placed on the heart muscle.
- It led to increased cerebrovascular resistance and breathing frequency, while simultaneously decreasing cerebral blood flow velocity.

Conclusion: *Ingesting Red Bull creates an overall negative hemodynamic profile, specifically stressing the heart and altering blood flow to the brain.*

Relevance: *Core evidence for both Impact 1 (Blood Pressure) and Impact 2 (Heart Rate). Directly links a specific, popular energy drink to the exact physiological changes being investigated in young adults.*

SOURCE 03

Caffeine as a Cardiovascular Protective Agent: A Comprehensive Review of Mechanisms and Outcomes

Authors: Ameer Awashra, Mohammed AbuBaha, et al.

PMID: 41782071 · **DOI:** 10.1186/s40001-026-04089-2

Link: [View on PubMed](#) →

KEY FINDINGS

- There is a non-linear, J-shaped association between caffeine intake and cardiovascular outcomes — moderate consumption often shows the lowest risk profile.
- Habitual caffeine use leads to tolerance, which attenuates (weakens) the acute blood pressure spikes seen in non-habitual users — explaining why responses vary widely between individuals.
- Cardiovascular responses vary significantly based on genetics, baseline blood pressure, and the form of caffeine consumed (e.g. unfiltered coffee vs. pure caffeine in energy drinks).

Conclusion: *While moderate intake is generally safe, the diversity of responses underscores the need to separate the effects of pure caffeine — as found in energy drinks — from other caffeinated beverages.*

Relevance: *Useful for adding depth to the discussion section. Explains why some young adults experience severe blood pressure spikes while others do not (tolerance and genetics). Also supports the alleviation section by establishing that moderate, controlled intake is generally safe.*

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